

Data Model Documentation

Overview

The employee leave application manages data related to employees, their leave requests, and the approval process within the organization. This data model ensures that all necessary information is captured and maintained to facilitate the leave management process efficiently. The system manages the following core domains:

- **Employees:** Responsible for storing employee details, their roles, and their leave balances.
- **Leave Requests:** Responsible for capturing leave requests submitted by employees, including the type of leave, duration, and status.
- **Approvals:** Responsible for tracking the approval status of leave requests, including who approved or rejected the request and when.

Entity Relationship Diagram

Below is an ASCII diagram representing the entities and their relationships:



Entities

Employee

Purpose: Represents an employee within the organization, including their personal details and leave balance.

Table/Collection Name: employees

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
name	String	Yes	Max 255 chars	Employee's full name
email	String	Yes	Max 255 chars, unique	Employee's email address
role	String	Yes	Max 100 chars	Employee's role within the organization
leave_balance	Integer	Yes	Min 0	Total leave days available
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many LeaveRequest (1:N): Inverse via LeaveRequest.employee_id
- Has one LeaveBalance (1:1): Inverse via LeaveBalance.employee_id

Validation Rules:

- name must be non-empty and within 255 characters
- email must be non-empty, unique, and a valid email format
- role must be non-empty and within 100 characters
- leave_balance must be a non-negative integer

LeaveRequest

Purpose: Captures details of a leave request submitted by an employee, including the type of leave, duration, and status.

Table/Collection Name: leave_requests

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to employees.id	Employee who submitted the request
leave_type	String	Yes	Max 100 chars	Type of leave (e.g., sick, vacation)
start_date	Date	Yes	Must be in the future	Start date of the leave
end_date	Date	Yes	Must be after start_date	End date of the leave
status	String	Yes	Must be one of (Pending, Approved, Rejected)	Current status of the request
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references employees.id
- Has one Approval (1:1): Inverse via Approval.leave_request_id

Validation Rules:

- leave_type must be non-empty and within 100 characters
- start_date and end_date must be valid dates and end_date must be after start_date
- status must be one of (Pending, Approved, Rejected)

LeaveBalance

Purpose: Tracks the leave balance for each employee, including the types of leave available.

Table/Collection Name: leave_balances

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
employee_id	UUID	Yes	Foreign key to employees.id	Employee associated with the balance
sick_leave	Integer	Yes	Min 0	Number of sick leave days remaining
vacation_leave	Integer	Yes	Min 0	Number of vacation leave days remaining
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Employee (N:1): Foreign key employee_id references employees.id

Validation Rules:

- sick_leave and vacation_leave must be non-negative integers

Approval

Purpose: Captures the approval status of a leave request, including the approver and the timestamp of the approval.

Table/Collection Name: approvals

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
leave_request_id	UUID	Yes	Foreign key to leave_requests.id	Leave request being approved

Field Name	Type	Required	Constraints	Description
approver_id	UUID	Yes	Foreign key to employees.id	Employee who approved the request
status	String	Yes	Must be one of (Approved, Rejected)	Approval status
comment	String	No	Max 500 chars	Comment from the approver
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to LeaveRequest (N:1): Foreign key leave_request_id references leave_requests.id
- Belongs to Employee (N:1): Foreign key approver_id references employees.id

Validation Rules:

- status must be one of (Approved, Rejected)
- comment must be within 500 characters if provided

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
leave_requests	employee_id	employees	id	N:1	CASCADE
leave_balances	employee_id	employees	id	N:1	CASCADE
approvals	leave_request_id	leave_requests	id	N:1	CASCADE
approvals	approver_id	employees	id	N:1	CASCADE

Unique Constraints

- `employees(email)`: Ensures no duplicate email addresses
- `leave_requests(employee_id, start_date, end_date)`: Ensures no overlapping leave requests for the same employee

Data Validation Rules

Business Rules:

- **Leave Balance Check:** Before approving a leave request, ensure the employee has sufficient leave balance for the requested leave type.
- **Date Range Validity:** Ensure the leave request's start and end dates do not overlap with existing leave requests for the same employee.
- **Approval Chain:** Ensure that only authorized employees can approve leave requests.

Format Rules:

- **Email Format:** `email` must match the regex
`^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$`
- **Leave Type:** `leave_type` must be one of (sick, vacation, personal)

Data Lifecycle

Creation

- A new `Employee` is created when a new employee is hired.
- Default `leave_balance` is set to 0 for both sick and vacation leave.
- `created_by` is automatically set to the HR manager creating the record.

Updates

- `Employee.leave_balance` can only be updated by the HR department.
- `updated_at` is automatically updated on any modification to an employee's details.

Archival / Soft Delete

- Records are not archived but marked with a `deleted_at` timestamp when an employee leaves the organization.
- Soft-deleted records are excluded from normal queries.

Indexes

- Primary key: id
- Foreign keys: employee_id, leave_request_id, approver_id
- Search fields: name, email, leave_type (for common queries)

Example Records

Employee

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "John Doe",
  "email": "john.doe@example.com",
  "role": "Software Engineer",
  "leave_balance": {
    "sick_leave": 10,
    "vacation_leave": 20
  },
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

LeaveRequest

```
{
  "id": "123e4567-e89b-12d3-a456-426614174000",
  "employee_id": "550e8400-e29b-41d4-a716-446655440000",
  "leave_type": "vacation",
  "start_date": "2026-02-01",
  "end_date": "2026-02-05",
  "status": "Pending",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

Approval

```
{
  "id": "789e0123-4567-89ab-cdef-0123456789ab",
  "leave_request_id": "123e4567-e89b-12d3-a456-426614174000",
  "approver_id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "Approved",
  "comment": "Leave approved for vacation",
  "created_at": "2026-01-16T09:00:00Z",
  "updated_at": "2026-01-16T09:00:00Z"
}
```